

1) Design a class hierarchy

CLASS Person

 Name

 Age

END CLASS

CLASS Student INHERITS Person

 Register Number

 Marks[5]

 METHOD CalculateAverage()

 Sum ← 0

 FOR i ← 0 TO 4

 Sum ← Sum + Marks[i]

 END FOR

 RETURN Sum / 5

 END METHOD

END CLASS

CLASS ResearchStudent INHERITS Student

 Research Area

 METHOD DisplayCompleteDetails()

 PRINT Name

 PRINT Age

 PRINT Register Number

 PRINT Marks

 PRINT "Average = ", CalculateAverage()

 PRINT Research Area

 END METHOD

END CLASS

BEGIN

CREATE ResearchStudent Rs

READ RS.Name

READ RS.Age

READ RS.Register Number

READ RS.Marks

READ RS.ResearchArea

Average $\leftarrow$ Rs.CalculateAverage()

RS.DisplayCompleteDetails()

END.

2) Complete the Missing pseudocode

BEGIN

READ N

READ ARRAY A

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow (i+1)$ TO $N-1$

IF $A[i] == A[j]$

DELETE $A[j]$

END IF

END FOR

END FOR

PRINT MODIFIED Array

END.

3) Debug the pseudoCode

CLASS Area

METHOD area(length)

RETURN length * length

END METHOD

METHOD area(length, breadth)

RETURN length * breadth

END METHOD

END CLASS

BEGIN

CREATE OBJECT A OF Area

READ length

PRINT "Area of Square =", A.area(length)

READ length, breadth

PRINT "Area of Rectangle =", A.area(length, breadth)

END.

4) Trace the Execution

Pseudo Code:

$$i = 1 \rightarrow \text{sum} = 0 - 1 = -1$$

$$i = 2 \rightarrow \text{sum} = -1 + 2 = 1$$

$$i = 3 \rightarrow \text{sum} = 1 - 3 = -2$$

$$i = 4 \rightarrow \text{sum} = -2 + 4 = 2$$

$$i = 5 \rightarrow \text{sum} = 2 - 5 = -3$$

Output:

-3.

5) Write Efficient Pseudo Code

BEGIN

READ N

READ first element

largest $\leftarrow$ first element

Second largest $\leftarrow -\infty$

FOR $i \in 2$ TO N

READ num

IF num > largest THEN

second largest $\leftarrow$ largest

largest $\leftarrow$ num

ELSEIF num > Second largest AND num $\neq$ largest THEN

second largest $\leftarrow$ num

ENDIF

ENDFOR

IF second largest $\neq -\infty$ THEN

PRINT "second largest not found"

ELSE

PRINT second Largest

ENDIF

END

6) Convert Problem Statement into Pseudocode (Interfaces)

INTERFACE Payment

METHOD pay (amount)

END INTERFACE

CLASS CreditCard IMPLEMENTS Payment

METHOD pay (amount)

PRINT "paid", amount, "using credit Card"

END METHOD

END CLASS

UPI

CLASS ~~Net Banking~~ IMPLEMENTS Payment

METHOD pay (amount)

PRINT "paid", amount, "using UPI"

END METHOD

END CLASS

CLASS Net Banking IMPLEMENTS Payment

METHOD pay (amount)

PRINT "paid", amount, "using Net Banking"

END METHOD

END CLASS

BEGIN

READ choice

READ amount

IF choice = 1 THEN

Payment ← CreditCard

ELSE IF choice = 2 THEN

Payment ← UPI

ELSE

Payment ← Net Banking

ENDIF

Payment. pay (amount)

END

7) Complete the PseudoCode

Filling Blanks:

CLASS Animal

METHOD speak()

PRINT "Animal"

END METHOD

END CLASS

CLASS Dog EXTENDS Animal

METHOD speak()

PRINT "Barks"

END METHOD

END CLASS

Animal obj ← NEW Dog()

obj.speak()

8) Find Logical Errors

Logical Mistakes:

CLASS Employee

-DATA

id

name

CONSTRUCTOR (id, name)

this.id ← id

this.name ← name

END CONSTRUCTOR

END CLASS.

9) Case Study (integrated problem)

CLASS vehicle

DATA

vehicle No

owner Name

CONSTRUCTOR (vehicle No, owner Name)

this.vehicle No ← vehicle No

this.owner Name ← owner Name

END CONSTRUCTOR

METHOD display()

PRINT vehicle No

PRINT owner Name

END METHOD

END CLASS

CLASS car EXTENDS vehicle

DATA

FuelType

CONSTRUCTOR (vehicle No, owner Name, fuelType)

CALL vehicle (vehicle No, owner Name)

this.fuelType ← fuelType

END CONSTRUCTOR

METHOD display()

CALL vehicle.display()

PRINT fuelType

END METHOD

END CLASS

CLASS TRUCK EXTENDS vehicle

DATA

load capacity

CONSTRUCTOR (vehicleNo, ownerName, loadCapacity)

CALL vehicle (vehicleNo, ownerName)

this.loadCapacity ← loadCapacity

END CONSTRUCTOR

METHOD display()

CALL vehicle.display()

PRINT loadCapacity

END METHOD

END CLASS

BEGIN

READ N

DECLARE vehicles[N]

FOR i ← 1 TO N

READ type

READ vehicleNo

READ ownerName

IF type = "car" THEN

READ fuelType

vehicles[i] ← NEW car (vehicleNo, ownerName, fuelType)

ELSEIF type = "truck" THEN

READ loadCapacity

vehicles[i] ← NEW Truck (vehicleNo, ownerName,
loadCapacity).

ENDIF

END FOR

FOR $i \leftarrow 1$ TO N

vehicles[i].display()

END FOR

END.